

## FORUM

# A “Limitless Frontier”? Neoliberalism, Silicon Valley, and the Commercialization of Space

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*Over the last forty years, analysts at American and British neoliberal think tanks have published a plethora of reports and commentaries celebrating the commercialization of space. This article examines why neoliberal think tanks have developed such a long-term idiosyncratic interest in space policies. I argue that space acts as an affective mobilizer that enables neoliberal ideologues to reactivate their historical ideological struggle against collectivism while simultaneously helping them to rescue the neoliberal project from numerous epistemological threats plaguing it. I first analyze how think tanks’ space activism operates in the context of their concerted struggle since the 1980s against “collectivist” international law. Second, I examine how they use the myth of the space frontier to revive the legitimization of private property as a core pillar of neoliberal thought. The last section analyzes how neoliberal think-tankers have come to offer space as the ideal solution to pressing environmental problems.*

For Services provided on Mars, or in transit to Mars via Starship or other spacecraft, the parties recognize Mars as a free planet and that no Earth-based government has authority or sovereignty over Martian activities. Accordingly, Disputes [*sic*] will be settled through self-governing principles, established in good faith, at the time of Martian settlement.

Starlink terms of service, 2022<sup>1</sup>

## Introduction

When registering with Starlink, Elon Musk’s satellite-based Internet network, new customers may be surprised to find the statement quoted above in the company’s terms of service. Considering that Starlink’s activities are currently limited to Earth and that the

<sup>1</sup>“Starlink Terms of Service,” Starlink, at [www.starlink.com/legal/documents/DOC-1020-91087-64](http://www.starlink.com/legal/documents/DOC-1020-91087-64) (accessed 20 April 2026).

legality of this contractual provision seems rather questionable, it is tempting to dismiss it as another one of Musk's empty provocations. Yet commentators within the galaxy of think tanks associated with the neoliberal thought collective have taken it seriously. While TechFreedom's counsel Corbin Barthold has defended Starlink against environmental regulators (including in front of a US Court of Appeals in 2021 and 2023) for the sake of Musk's extraplanetary ambitions, the Foundation for Economic Education's Thomas Walker-Werth praised Musk's project of a stateless Mars.<sup>2</sup> Asking his readers to "imagine" the benefits opened up by the emergence of outer-space territories freed from earthly states' regulations and restrictions on innovation and entrepreneurship, Walker-Werth speculated that "Mars, asteroids, and the Jovian and Saturnian moons ... could all come alive with a plethora of different productive settlements, each with the potential to blossom into a new civilization."<sup>3</sup>

This defense of Musk's Starlink initiative is not an isolated instance of neoliberal think tanks' incursion into interplanetary politics. American neoliberal think tanks have sustained interest in space topics for decades, and their understanding of space often converges with the one promoted by Silicon Valley tycoons like Elon Musk, Jeff Bezos, and Peter Thiel. Over the last forty years, the American Enterprise Institute, the Cato Institute, the Competitive Enterprise Institute (CEI), the Foundation for Economic Education (FEE), the Heartland Institute, the Heritage Foundation, the Reason Foundation, TechFreedom, and other groups have published a plethora of reports, articles, and commentaries on the commercialization of space resources and territory.<sup>4</sup> Scholars associated with the neoliberal thought collective, such as Walter Block or 2007 Hayek Prize laureate Peter Leeson, have also supported privatizing

<sup>2</sup>Corbin Barthold, "Rival Wants Regulators to Cripple Elon Musk's Satellite Project: Starlink's Fight for the Future," *Bulwark*, 3 Aug. 2021, at [www.thebulwark.com/rival-wants-regulators-to-cripple-elon-musks-satellite-project](http://www.thebulwark.com/rival-wants-regulators-to-cripple-elon-musks-satellite-project); Corbin Barthold, James E. Dunstan, and Berin Szóka, "Brief of Amicus Curiae TechFreedom in Support of Appellee and Affirmance on Appeal from the Federal Communications Commission, IBFS File Nos. SAT-LOA-20200526-00055 and SAT-AMD-20210818," 20 June 2023, at <https://techfreedom.org/wp-content/uploads/2023/06/TF-22-1337-International-Dark-Sky-Association-Inc.-v.-FCC.pdf>; Thomas Walker-Werth, "Elon Musk Is Right: Governments Shouldn't Control Land on Other Planets," *FEE Stories*, 26 Jan. 2022, at <https://fee.org/articles/elon-musk-is-right-governments-shouldn-t-control-land-on-other-planets>.

<sup>3</sup>Walker-Werth, "Elon Musk Is Right."

<sup>4</sup>To mention only a few examples: Edward L. Hudgins, ed., *Space: The Free-Market Frontier* (Washington, DC, 2002); Rand Simberg, "Homesteading the Final Frontier" (Competitive Enterprise Institute, Feb. 2012), at <https://cei.org/studies/homesteading-the-final-frontier>; TechFreedom, "Moon Express Reveals Need for Space Law," *TechFreedom*, June 2016, at <https://techfreedom.org/moon-express-reveals-need-for-space-law>; Christopher Talgo, "A New Era of American Space Exploration," Heartland Institute, Jan. 2018, at [www.heartland.org/news-opinion/news/a-new-era-of-american-space-exploration](http://www.heartland.org/news-opinion/news/a-new-era-of-american-space-exploration); Jeff Greason and James C. Bennett, "The Economics of Space: An Industry Ready to Launch" (Reason Foundation, June 2019), at <https://reason.org/policy-study/the-economics-of-space>; Dean Cheng, "Outer Space and Private Property," Heritage Foundation, Sept. 2020, at [www.heritage.org/space-policy/commentary/outer-space-and-private-property](http://www.heritage.org/space-policy/commentary/outer-space-and-private-property); Saul Zimet, "What Elon Musk's Critics Get Wrong about Colonizing Mars," *FEE Stories*, May 2021, at <https://fee.org/articles/what-elon-musk-s-critics-get-wrong-about-colonizing-mars>; James Pethokoukis, "My God, It's Full of Stars!," American Enterprise Institute, July 2022, at [www.aei.org/articles/my-god-its-full-of-stars](http://www.aei.org/articles/my-god-its-full-of-stars).

space.<sup>5</sup> British neoliberal think tanks have followed a similar trend, demonstrated by the work of Andrew Lilico (formerly of Policy Exchange) and Mark Littlewood (the Institute of Economic Affairs—IEA) in the *Daily Telegraph* and *The Times*, and Rebecca Lowe's report on property rights on the Moon for the Adam Smith Institute (ASI).<sup>6</sup> Lowe and Lilico have also expounded their views on space policies in *CapX*, a publication platform attached to the Centre for Policy Studies.<sup>7</sup>

The aim of this article is to understand why neoliberal think tanks have developed this long-term idiosyncratic interest in space policies. I demonstrate that—as a realm beyond earthly limits and current institutional boundaries, yet seemingly within the reach of capital—space has become a locus for the projection of neoliberal fantasies. I argue that space acts as an affective mobilizer that enables neoliberal ideologues to reactivate their historical ideological struggle against collectivism while simultaneously helping them to rescue the neoliberal project from numerous epistemological threats—including the theoretical problem of private property's origins and the concrete danger of climate change.

The article thus intervenes in the rich and still-expanding scholarship on the history of neoliberalism—here understood as a tradition of thought that spans from the inter-war period until the present day, and which has been articulated by a well-organized international intellectual movement centering on the Mont Pèlerin Society (MPS) and a sprawling ecosystem of free-market think tanks.<sup>8</sup> In their pioneering edited volume *The Road from Mont Pèlerin* (2009), Philip Mirowski and Dieter Plehwe show that the select group of scholars, journalists, corporate leaders, politicians, and think-tankers that took part in MPS meetings formed a “thought collective.”<sup>9</sup> Through their discussions and further organization, they collectively and iteratively shaped a common neoliberal identity and worked together to advance a common neoliberal cause.<sup>10</sup> Around Plehwe, a subfield in neoliberal studies has focused on how free-market think

<sup>5</sup>Peter Lothian Nelson and Walter E. Block, *Space Capitalism: How Humans Will Colonize Planets, Moons, and Asteroids* (Cham, 2018); Alexander William Salter and Peter T. Leeson, “Celestial Anarchy,” *Cato Journal* 34/3 (2014), 581–96. The Hayek Prize is awarded by the Fund for the Study of Spontaneous Orders, supported by the Atlas Economic Research Foundation.

<sup>6</sup>Rebecca Lowe, “Space Invaders: Property Rights on the Moon” (Adam Smith Institute, 2022); Andrew Lilico, “Let’s Turn the Red Planet Green,” *Times Live*, 20 Aug. 2014, at [www.timeslive.co.za/news/south-africa/2014-08-20-lets-turn-the-red-planet-green](http://www.timeslive.co.za/news/south-africa/2014-08-20-lets-turn-the-red-planet-green); Mark Littlewood, “Space Offers True Spirit of Enterprise That the Critics Should Not Dismiss,” *The Times*, 18 Oct. 2021, at [www.thetimes.co.uk/article/space-offers-true-spirit-of-enterprise-that-the-critics-should-not-dismiss-nncwq0mjt](http://www.thetimes.co.uk/article/space-offers-true-spirit-of-enterprise-that-the-critics-should-not-dismiss-nncwq0mjt).

<sup>7</sup>Lilico, “Is Colonising Space Economically Viable?,” *CAPX*, May 2017, at <https://capx.co/is-colonising-space-economically-viable>; Lowe, “Shoot for the Moon: How to Give Humanity a Stake in Space,” *CAPX*, Feb. 2022, <https://capx.co/shoot-for-the-moon-how-to-give-humanity-a-stake-in-space>.

<sup>8</sup>For historical studies on the neoliberal movement see Philip Mirowski and Dieter Plehwe, eds., *The Road from Mont Pèlerin: The Making of the Neoliberal Thought Collective* (Cambridge, MA, 2009); Daniel Stedman Jones, *Masters of the Universe: Hayek, Friedman, and the Birth of Neoliberal Politics* (Princeton, 2012); Angus Burgin, *The Great Persuasion: Reinventing Free Markets since the Depression* (Cambridge, MA, 2013); Quinn Slobodian, *Globalists: The End of Empire and the Birth of Neoliberalism* (London, 2018); or Dieter Plehwe, Quinn Slobodian, and Philip Mirowski, eds., *Nine Lives of Neoliberalism* (London, 2020).

<sup>9</sup>Mirowski and Plehwe, *The Road from Mont Pèlerin*.

<sup>10</sup>Dieter Plehwe, “Introduction,” in Mirowski and Plehwe, *The Road from Mont Pèlerin*, 1–42; see also Ben Jackson, “Putting Neoliberalism in Its Place,” *Modern Intellectual History* 19/3 (2022), 982–95, at 983.

tanks orbiting the MPS have specifically contributed to this collective endeavor.<sup>11</sup> Marie-Laure Djelic and Reza Mousavi, for instance, examine the Atlas Network—to which all the think tanks discussed in this article are affiliated—as “the hub organization of a dense transnational community of neoliberal think tanks” that aims to diffuse and embed neoliberal ideas globally.<sup>12</sup> These recent studies on neoliberal think tanks have thus concentrated on how they influence policy makers and institutions, as well as on specific areas of neoliberal think-tank activism, like climate change policy skepticism.<sup>13</sup> Yet space policy as an area of neoliberal think-tank intervention has often been neglected despite explicit crossovers. For instance, I demonstrate here that space plays an important part in the alternative proposals that neoliberal think-tankers have put forward to tackle climate change.

Conversely, with the publicity given to Musk’s Mars colonization plans, there is a burgeoning scholarship conceptualizing the current wave of (commercial) space expansionism, to highlight either its colonial underpinnings (as in the works of Alina Utrata and Mary-Jane Rubenstein) or its questioning of international space law.<sup>14</sup> Yet these studies tend to focus on corporations, like SpaceX, somewhat overlooking the role of organized advocacy networks that have been long-term supporters of space colonization.

The specific contribution of this article is to bring these two scholarly traditions together by considering the involvement of neoliberal think tanks in the current push for further commercializing space. I am not the first, however, to tackle this issue. Matthew Johnson’s excellent 2020 doctoral thesis (recently turned into a book), “Mining the High Frontier,” discusses the emergence of New Space advocacy networks (the umbrella term for the private space industry) since the 1920s and their

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<sup>11</sup>For a few examples, see Christian E. Weller and Laura Singleton, “Peddling Reform: The Role of Think Tanks in Shaping the Neoliberal Policy Agenda for the World Bank and International Monetary Fund,” in Dieter Plehwe, Bernhard J. A. Walpen, and Gisela Neunhöffer, eds., *Neoliberal Hegemony: A Global Critique* (Abingdon, 2006), 70–86; Plehwe, “Introduction”; Karin Fischer and Dieter Plehwe, “Neoliberal Think Tank Networks in Latin America and Europe: Strategic Replication and Cross-national Organizing,” in Alejandra Salas-Porras and Georgina Murray, eds., *Think Tanks and Global Politics* (New York, 2017), 159–86; Marie-Laure Djelic and Reza Mousavi, “How the Neoliberal Think Tank Went Global: The Atlas Network, 1981 to the Present,” in Plehwe, Slobodian, and Mirowski, *Nine Lives of Neoliberalism*, 257–83.

<sup>12</sup>Djelic and Mousavi, “How the Neoliberal Think Tank Went Global,” 258. For a list of Atlas’s partners see “Partner Directory,” Atlas Network, archived 13 June 2021 at <https://web.archive.org/web/20210613144830/https://www.atlasnetwork.org/partners/global-directory>.

<sup>13</sup>For the former see, for instance, Weller and Singleton, “Peddling Reform”; Dorothee Bohle and Gisela Neunhöffer, “Why Is There No Third Way? The Role of Neoliberal Ideology, Networks and Think Tanks in Combating Market Socialism and Shaping Transformation in Poland,” in Plehwe, Walpen, and Neunhöffer, *Neoliberal Hegemony*, 89–104. For the latter see Dieter Plehwe, “Think Tank Networks and the Knowledge–Interest Nexus: The Case of Climate Change,” *Critical Policy Studies* 8/1 (2014), 101–15.

<sup>14</sup>For the former see Alina Utrata, “Engineering Territory: Space and Colonies in Silicon Valley,” *American Political Science Review* 118/3 (2024), 1097–1109; and Mary-Jane Rubenstein, *Astrotopia: The Dangerous Religion of the Corporate Space Race* (London, 2022). For the latter see, for example, Marco A. Janssen and Xiao-Shan Yap, “Governing Outer Space as a Commons Is Critical for Addressing Commons on Earth,” *International Journal of the Commons* 18/1 (2024), 33–8; as well as Aaron Boley and Michael Byers, “U.S. Policy Puts the Safe Development of Space at Risk,” *Science* 370/6513 (2020), 174–5.

interplay with neoliberal policy networks.<sup>15</sup> While frequently drawing on Johnson's work, my article is less about discussing the historical partnership between New Space and neoliberal advocacy groups, than about understanding *why* space still has such a power of attraction on today's neoliberal ideologues and scholars.

To explore this, I mainly focus on US and UK think tanks' post-2000 publications—that is, on contributions written after the founding of Jeff Bezos's Blue Origin, Elon Musk's SpaceX and Richard Branson's Virgin Galactic in 2000, 2002, and 2004 respectively, which marks a new chapter in the history of New Space. This corpus can be divided into two broad categories. On the one hand, policy-oriented documents generally written on behalf of neoliberal think tanks by engineers and space entrepreneurs (e.g. Rand Simberg for the Competitive Enterprise Institute, Jeff Greason and James E. Bennett for the Reason Foundation), legal counsels (e.g. Corbin Barthold, James E. Dunstan and Berin Szóka for TechFreedom), or researchers (Rebecca Lowe for the Adam Smith Institute) tend to make room for some level of state intervention, for example by assuming that the state will build infrastructure to facilitate private space colonization.<sup>16</sup> On the other hand, more abrasive and frankly unhinged opinion pieces, often connected with the libertarian branches of the neoliberal think-tank network—for instance, think tanks like FEE (e.g. through its Hazlitt fellows, Saul Zimet and Thomas Walker-Werth) and the Heartland Foundation (e.g. Heartland Policy adviser and climate change denier Charles N. Steele), and controversial figures like Block and Lilico—offer anti-state accelerationist arguments.

Before going further, it is important to recognize that part of the scholarship differentiates libertarianism from neoliberalism, often to highlight the incorporation of libertarianism's right-wing factions into today's alt-right coalition and its opposition to more globalist forms of neoliberalism.<sup>17</sup> While acknowledging their difference and particularity (for instance by being attentive to the inflection specific to libertarianism, such as its virulent antistatism or its focus on homesteading), I prefer instead to follow the steps of Melinda Cooper and Quinn Slobodian in emphasizing their

<sup>15</sup>Matthew Johnson, "Mining the High Frontier: Sovereignty, Property and Humankind's Common Heritage in Outer Space" (Ph.D. thesis, University of Technology, Sydney, 2020); Johnson, *Property Rights in Outer Space: Mining, Techno-utopian Imaginaries, and the Privatisation of the Off-World Frontier* (Abingdon, 2024).

<sup>16</sup>Simberg, "Homesteading the Final Frontier"; Greason and Bennett, "Economics of Space"; James E. Dunstan and Berin Szóka, "Written Testimony of James E. Dunstan and Berin Szóka: Senate Committee on Commerce, Science, & Transportation Subcommittee on Space, Science, and Competitiveness," May 2017, at [www.commerce.senate.gov/services/files/A9AD88B2-9636-4291-A5B0-38BC0FF6DA90](http://www.commerce.senate.gov/services/files/A9AD88B2-9636-4291-A5B0-38BC0FF6DA90). James E. Dunstan, "Comments of TechFreedom James E. Dunstan in the Matter of Moon to Mars Objectives," June 2022, at <https://techfreedom.org/wp-content/uploads/2022/06/TechFreedom-Comment-Moon-to-Mars-6-3-22.pdf>; Lowe, "Space Invaders."

<sup>17</sup>See, for instance, Jacob Jensen, "Repurposing Mises: Murray Rothbard and the Birth of Anarchocapitalism," *Journal of the History of Ideas* 83/2 (2022), 315–32; and Daniel Bessner, "Murray Rothbard, Political Strategy, and the Making of Modern Libertarianism," *Intellectual History Review* 24/4 (2014), 441–56. For a study of paleolibertarianism's connections with the alt-right see Quinn Slobodian, "The Anti-'68ers and the Racist-Libertarian Alliance: How a Schism among Austrian School Neoliberals Helped Spawn the Alt Right," *Cultural Politics* 15/3 (2019), 371–86; and Melinda Cooper, "The Alt-Right: Neoliberalism, Libertarianism and the Fascist Temptation," *Theory, Culture and Society* 38/6 (2021), 29–50.

common belonging to the neoliberal thought collective.<sup>18</sup> To extend Cooper's words, libertarianism—in both its reformist and its radical branches—sits “within the broad constellation of American neoliberalisms,” as is well illustrated by the trajectory of paleolibertarianism's figurehead, Murray Rothbard.<sup>19</sup> Rothbard led the revival of neoliberal Austrian economics in the US and was a long-time member of the MPS—and this despite clear tensions within the thought collective on the issue of culture and race.<sup>20</sup> Other figures of American libertarianism mentioned in the article—like Walter Block and Robert Poole—similarly still hold membership in the MPS, further justifying my decision to keep neoliberalism as the common denominator between the free-market think tanks studied in this article.<sup>21</sup>

My argument proceeds in three stages. First, I examine how the neoliberal publications on space operate in the context of their historical fight against international law: I go back to the first emergence of these publications in the late 1970s to then focus on their most recent institutional victories. The second and third parts of the article focus on post-2000 neoliberal think tanks' publications on space by adopting a more thematic approach. The second part analyzes how space functions as a fantasy in neoliberal discourse that allows think-tankers to wish away a number of epistemological issues putting their ideal society at risk. The last section focuses on how space is presented as the ideal solution to the most pressing environmental threats and concludes with a discussion of the ideological role of the entrepreneur.

### **Appropriating the last commons: the battle against the Outer Space Treaty *The Moon Agreement and the specter of “collectivism” in the 1980s***

While I mainly concentrate on post-2000 publications, it is useful to contextualize the emergence of space as a topic of discussion among neoliberal think-tankers. The first publications dedicated to the commercialization of space found in the online archives of US and UK neoliberal think tanks date from the late 1970s. They include a series of articles published by *Reason* magazine (from which sprang the Reason Foundation) between 1978 and 1985, followed by a multiplication of reports from FEE, the Cato Institute, the Heritage Foundation and the American Enterprise Institute after 1983.<sup>22</sup>

<sup>18</sup> Cooper, “The Alt-Right”; Quinn Slobodian, *Hayek's Bastards: The Neoliberal Roots of the Populist Right* (London, 2025).

<sup>19</sup> Cooper, “The Alt-Right,” 30. For an account of the split within libertarianism see Bessner, “Murray Rothbard.”

<sup>20</sup> Slobodian, *Hayek's Bastards*, 39.

<sup>21</sup> The same can be said of other libertarian figures not mentioned in the article, like Charles Koch and Ed Crane, who cofounded the Cato Institute with Rothbard in 1977, before ousting him in 1981. See “Mont Pelerin Society Directory 2013,” retrieved by DeSmog, 2013, at [www.desmog.com/wp-content/uploads/files/MPS-2013-membership-listing\\_Redacted.pdf](http://www.desmog.com/wp-content/uploads/files/MPS-2013-membership-listing_Redacted.pdf).

<sup>22</sup> For *Reason* publications see Robert Poole, “African Deception,” *Reason*, July 1978, at <https://reason.com/1978/07/01/african-deception>; Poole, “Free Space!”, *Reason*, April 1979, at <https://reason.com/1979/04/01/free-space>; Poole, “Space Piracy,” *Reason*, Jan. 1980, at <https://reason.com/1980/01/01/space-piracy>; Peter Vajk, “No More Doomsday: ‘Limits to Growth’ Is a Myth—But There Are Still Plenty of Good Reasons for Going into Space,” *Reason*, April 1979, at <https://reason.com/1979/04/01/no-more-doomsday>; James C. Bennett, “The Second Space Race,” *Reason*, Nov. 1981, at



This trend coincides with Reagan's two presidential mandates marked by his commitment to develop further the commercial use of space, as, for instance, illustrated by the launch in 1984 of his Commercial Space Launch Act.<sup>23</sup> In the same way as post-2000 neoliberal think tanks' publications on space have been galvanized by the founding of Jeff Bezos's Blue Origin, Elon Musk's SpaceX, and Richard Branson's Virgin Galactic, the 1980s reports accompany a rapid development of the commercial launch industry for communication satellites—with, for instance, the creation of Arianespace, the world's first commercial space transportation company, in 1980, and the launch of Space Services Inc.'s *Conestoga 1*, the first privately funded spaceflight to reach space, in 1982.<sup>24</sup> One recurrent theme of these early publications is a critique of the alleged monopoly of the US National Aeronautics and Space Administration (NASA) on space transportation and its space shuttle program.<sup>25</sup> Instead, these articles argue that only the private sector can legitimately lead the commercial development of space.

For the contributors of the 1980s, rapid technological developments in the space industry, and space telecommunication in particular, called for the introduction of private-property rights in an environment where they did not yet (and still do not) exist.<sup>26</sup> While the authors of these articles rarely give their theoretical references as they write for a nonspecialist readership or for policy makers, a 1987 *Cato Journal* article by Joel D. Scheraga stands out when it comes to the question of private property.<sup>27</sup> Scheraga used Ronald Coase's work on the distribution of property rights for radio and television frequencies to argue that the failure to assign property rights to scarce space resources, like geosynchronous orbits, would inevitably lead to an inefficient use of these resources—thus threatening, for instance, to impede future communications via

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<https://reason.com/1981/11/01/the-second-space-race>; H. Keith Henson and Arel Lucas, "Star Laws," *Reason*, Aug. 1982, at <https://reason.com/1982/08/01/star-laws>; and Patrick Cox, "Space Entrepreneurs: Their Business Is Rockets. Their Dream Is Space. Their Bane Is NASA," *Reason*, Jan. 1985, at <https://reason.com/1985/01/01/space-entrepreneurs>. See also Gary McGath, "Free Enterprise in Space" (Foundation for Economic Education, 1 March 1983), at <https://fee.org/articles/free-enterprise-in-space>; Paul R. Portney and Molly K. Macauley, "Slicing the Geostationary Pie: Property Rights in Orbit" (American Enterprise Institute, 26 July 1984), at [www.aei.org/articles/slicing-the-geostationary-pie-property-rights-in-orbit](http://www.aei.org/articles/slicing-the-geostationary-pie-property-rights-in-orbit); Milton R. Copulos, "The Perils of a NASA Space Monopoly" (Heritage Foundation, 20 June 1984), at [www.heritage.org/report/the-perils-nasa-space-monopoly](http://www.heritage.org/report/the-perils-nasa-space-monopoly); Milton R. Copulos, "Enterprise in Orbit: The Commercial Potential of Space" (Heritage Foundation, 20 November 1984), at [www.heritage.org/space-policy/report/enterprise-orbit-the-commercial-potential-space](http://www.heritage.org/space-policy/report/enterprise-orbit-the-commercial-potential-space); Copulos, "Government Obstacles to the Commercial Use of Space," Heritage Foundation backgrounder (4 March 1985), at [www.heritage.org/government-regulation/report/government-obstacles-the-commercial-use-space](http://www.heritage.org/government-regulation/report/government-obstacles-the-commercial-use-space); and Juliana Geran Pilon, "How the U.N. Is Off Course in Outer Space," Heritage Foundation backgrounder (8 Feb. 1985), at [www.heritage.org/global-politics/report/how-the-un-course-outer-space](http://www.heritage.org/global-politics/report/how-the-un-course-outer-space).

<sup>23</sup> Alan Pell Crawford, "An 'Industrial Policy' for Space?," *Policy Analysis* (Cato Institute, 25 April 1986), at [www.cato.org/policy-analysis/industrial-policy-space](http://www.cato.org/policy-analysis/industrial-policy-space); Edward L. Hudgins, "Next Step for U.S. Space Policy" (Heritage Foundation, 23 February 1988), at [www.heritage.org/space-policy/report/next-step-us-space-policy](http://www.heritage.org/space-policy/report/next-step-us-space-policy).

<sup>24</sup> Bennett, "Second Space Race"; McGath, "Free Enterprise"; Copulos, "Enterprise in Orbit"; Cox, "Space Entrepreneurs."

<sup>25</sup> Poole, "Free Space!"; Bennett, "Second Space Race"; Copulos, "The Perils"; Cox, "Space Entrepreneurs."

<sup>26</sup> Portney and Macauley, "Slicing the Geostationary Pie."

<sup>27</sup> Joel D. Scheraga, "Establishing Property Rights in Outer Space," *Cato Journal* 6/3 (1987), 889–903.

satellite.<sup>28</sup> He also alluded to Garrett Hardin's 1968 theory of the "tragedy of the commons" by arguing that the absence of defined property rights would lead to an overuse of resources—taking the near-extinction of Great Plains bison as his example.<sup>29</sup> Lastly, Scheraga referred to Harold Demsetz's framework to reassert his conviction that states would soon have a direct interest in establishing property rights in space as the benefits for doing so started to outweigh their social cost; states needed to leave it to the market as the market would "ensure that property rights to the scarce orbital slots [would be] defined and enforced."<sup>30</sup> Scheraga was here directly challenging the nascent international legal regime on space, and nominally the 1967 Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space (or "Outer Space Treaty"—OST), whose "failure to establish property rights" he judged to be "critical."<sup>31</sup>

Scheraga was far from isolated in his attack on the OST: most of the 1980s neoliberal think tanks' publications on space are a direct response and challenge to it and other international space legislation. They specifically intervene in the context of a lobbying campaign against the 1979 Agreement Governing the Activities of States on the Moon and Other Celestial Bodies (or "Moon Agreement") orchestrated by the L-5 Society (now the National Space Society), a grassroots movement of space enthusiasts and early representatives of the forming New Space lobby defending the interests of private space actors.<sup>32</sup> As a result of their coordinated efforts, the US never signed or ratified the agreement. Crucially, neoliberal networks supported this successful campaign. In January 1980, only a couple of weeks after the Moon Agreement was opened for signature, the Reason Foundation's cofounder Robert Poole exhorted his *Reason* readers to answer the L-5 Society's call for action in order not to let "the promise of space be drowned in a sea of socialism."<sup>33</sup> *Reason* followed up two years later by publishing an article by the L-5 Society's representatives, including its cofounder, Keith Henson, where they attacked the entire international legislation on space.<sup>34</sup>

The Moon Agreement is the last of a series of five United Nations treaties, around which the international space regime is organized.<sup>35</sup> Significantly, the other founding treaties and specifically its main one, the OST, had not—until then—attracted the neoliberal think tanks' attention. Of course, only a handful of the think tanks discussed

<sup>28</sup>Ibid., 893.

<sup>29</sup>Ibid., 893. Scheraga does not directly reference Hardin, quoting instead Deirdre McCloskey's 1985 monograph *The Applied Theory of Price*.

<sup>30</sup>Scheraga, "Establishing Property Rights in Outer Space," 894.

<sup>31</sup>Ibid., 392.

<sup>32</sup>Johnson, "Mining the High Frontier," 53–8.

<sup>33</sup>Poole, "Space Piracy."

<sup>34</sup>Henson and Lucas, "Star Laws."

<sup>35</sup>"Status of International Agreements Relating to Activities in Outer Space as at 1 January 2024," Committee on the Peaceful Uses of Outer Space, April 2024, at [www.unoosa.org/res/oosadoc/data/documents/2024/aac\\_105c\\_22024crp/aac\\_105c\\_22024crp\\_3\\_0\\_html/AC105\\_C2\\_2024\\_CRP03E.pdf](http://www.unoosa.org/res/oosadoc/data/documents/2024/aac_105c_22024crp/aac_105c_22024crp_3_0_html/AC105_C2_2024_CRP03E.pdf). The other agreements are the 1967 Outer Space Treaty, the 1968 Rescue Agreement, the 1972 Liability Convention, and the 1975 Registration Convention. To access the Moon Agreement see United Nations for Outer Space Affairs (UNOOSA), "Agreement Governing the Activities of States on the Moon and Other Celestial Bodies" (Moon Agreement), 5 Dec. 1979, available at [www.unoosa.org/oosa/en/ourwork/spacelaw/treaties/moon-agreement.html](http://www.unoosa.org/oosa/en/ourwork/spacelaw/treaties/moon-agreement.html).



here already existed prior to 1967—with the bulk of them created in the 1970s and 1980s. Yet what triggered their reaction is that they identified the Moon Agreement as bearing the imprint of the New International Economic Order (NIEO), promoted from 1974 by the Group of 77 representing developing and nonaligned countries at the United Nations.<sup>36</sup> The 1980s neoliberal think-tankers specifically opposed the agreement's definition of the Moon as the "heritage of all Mankind," which they argued "assumes that the resources and wealth developed by the industrial nations somehow are 'the common heritage' of all mankind, and as such should be transferred by right to the developing states of the Third World."<sup>37</sup> Such "unearned" transfer would be nothing less than "piracy," according to Robert Poole. Similarly, think-tankers like Poole condemned the agreement's provisions for the creation of an international "authority" (likely "controlled by a one-nation/one-vote assembly, whose decisions would be exempt from judicial review") to centrally plan the exploitation of space resources.<sup>38</sup> Their rejection of the Moon Agreement was reinforced by the campaign they ran simultaneously against the 1982 Law of the Sea Convention (LOSC), which also uses the concept of "common heritage of mankind."<sup>39</sup> Neoliberals saw both legal texts as outgrowth of the NIEO "doctrine" and its alleged socialist redistributive principles.<sup>40</sup> Their ideological opposition to the consecration of outer space and Earth's deep seabed as humanity's commons is well captured by the trajectory of Doug Bandow, a senior fellow at Cato and prior to this at Heritage, who also served as Reagan's deputy representative to the Third UN Conference on the Law of the Sea.<sup>41</sup> Bandow has zealously attacked the LOSC and the Moon Agreement since the 1980s, writing in 2020 that the latter was inspired by a group "represent[ing] largely socialist dictatorships which sought to guilt the West into transferring vast resources to their treasuries."<sup>42</sup>

Prior treaties, like the 1967 OST or the 1972 Liability Convention, were then a posteriori reinterpreted by the neoliberal think-tankers in light of the Moon Agreement.<sup>43</sup> Cato's Edward L. Hudgins was characteristically able to write in 1998 that both treaties

<sup>36</sup> Geran Pilon, "U.N. Is Off Course"; Copulos, "Government Obstacles." For two accounts of this period see Quinn Slobodian, *Globalists*, 218–62; and Jessica Whyte, *The Morals of the Market: Human Rights and the Rise of Neoliberalism* (London, 2019), 218–26.

<sup>37</sup> Geran Pilon, "U.N. Is Off Course." See also Copulos, "Government Obstacles."

<sup>38</sup> Poole, "Space Piracy."

<sup>39</sup> Florian Rabitz, "Space Resources and the Politics of International Regime Formation," *International Journal of the Commons* 17/1 (2023), 243–55.

<sup>40</sup> Geran Pilon, "U.N. Is Off Course."

<sup>41</sup> Doug Bandow, "Do Not Endorse the Law of the Sea Treaty," Cato Institute foreign-policy briefing (27 Jan. 1994), at [www.cato.org/sites/cato.org/files/pubs/pdf/fpb029.pdf](http://www.cato.org/sites/cato.org/files/pubs/pdf/fpb029.pdf).

<sup>42</sup> Doug Bandow, "Trump EO: The Moon and Other Celestial Bodies Should Be Open to Private Resource Development," *Cato Institute*, 28 April 2020, at [www.cato.org/publications/commentary/trump-eo-moon-other-celestial-bodies-should-be-open-private-resource](http://www.cato.org/publications/commentary/trump-eo-moon-other-celestial-bodies-should-be-open-private-resource); Bandow, "The Law of the Sea Treaty: Still Flawed and Increasingly Irrelevant," Heritage Foundation, 15 Jan. 1988, at [www.heritage.org/report/the-law-the-sea-treaty-still-flawed-and-increasingly-irrelevant](http://www.heritage.org/report/the-law-the-sea-treaty-still-flawed-and-increasingly-irrelevant); Bandow, "Washington's Night of the Living Dead: The Law of the Sea Treaty Stirs," *Forbes*, 12 Sept. 2011, at [www.cato.org/commentary/washingtons-night-living-dead-law-sea-treaty-stirs](http://www.cato.org/commentary/washingtons-night-living-dead-law-sea-treaty-stirs); Bandow, "Do Not Endorse"; Johnson, "Mining the High Frontier," 57.

<sup>43</sup> For instance, Bennett, "Second Space Race."

were “rather blatant forms of socialism.”<sup>44</sup> In other words, by 2000 the international space regime came to symbolize the remnants of the mid-century collectivist world order that the neoliberal thought collective had opposed in its quest for hegemony.<sup>45</sup> The think tanks’ coordinated attack on the outer-space regime thus offered the neoliberal thought collective a renewed opportunity to mobilize against a common enemy—a role that it still plays.

### *The post-2000 neoliberal critique of international space regime*

What is striking is that the neoliberal think tanks’ rhetoric on space legislation and entrepreneurship has remained remarkably similar up to the present. Some of the contributors writing against the international space regime in the 1980s, such as Doug Badow or space entrepreneur and *Reason* contributor James C. Bennett (who also served on Reagan’s 1984 Task Force on Space Commercialization) remain active, still authoring reports and commentaries for today’s neoliberal think tanks.<sup>46</sup>

The Moon Agreement’s common-heritage-based regime is characteristically still decried, for instance by TechFreedom’s counsel James Dunstan, as instituting a form of collective property of space resources and its provision for the creation of an “international regime” to manage and redistribute these resources as a form of intolerable taxation of mining activities in space.<sup>47</sup> To the great satisfaction of neoliberal think-tankers, only seventeen countries and no spacefaring nations have ratified it to this day—making it effectively, in the words of Bennett and his 2019 coauthor Jeff Greason, a “dead letter.”<sup>48</sup>

Neoliberals criticized the OST along similar lines, although they deemed it slightly more acceptable. Defining space as “the province of all mankind,” the OST guarantees free exploration and use of space by all states, as well as “free access to all areas of celestial bodies.”<sup>49</sup> Article II also stipulates that “outer space is not subject to national appropriation by claim of sovereignty, by means of use or occupation, or by any other means.” As states traditionally remain the legal guarantors of individuals’ private property rights, the treaty prevents appropriation of space resources and land, which makes it—as discussed further below—particularly objectionable to neoliberals. They could

<sup>44</sup>Edward L. Hudgins, “Martian Law,” Cato Institute, 15 Aug. 1998, at [www.cato.org/white-paper/martian-law](http://www.cato.org/white-paper/martian-law).

<sup>45</sup>Plehwe, “Introduction,” 5–6.

<sup>46</sup>Bennett, “Second Space Race”; Greason and Bennett, “Economics of Space.” For a short biography of Bennett see Reason Foundation, “James C. Bennett,” 2024, at <https://reason.org/author/james-c-bennett>.

<sup>47</sup>James E. Dunstan, “Can We (Legally) Colonize Space?” *Freethink*, 5 Sept. 2021, at [www.freethink.com/space/can-we-legally-colonize-space](http://www.freethink.com/space/can-we-legally-colonize-space). See also UNOOSA, “Agreement Governing the Activities of States on the Moon and Other Celestial Bodies.”

<sup>48</sup>Greason and Bennett, “Economics of Space,” 74; Simberg, “Homesteading the Final Frontier,” 7. These are the April 2024 numbers published by UN COPUOS; see Committee on the Peaceful Uses of Outer Space, “Status of International Agreements.”

<sup>49</sup>“Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies” (Outer Space Treaty), United Nations Office for Outer Space Affairs, 19 December 1966, accessible at [www.unoosa.org/oosa/en/ourwork/spacelaw/treaties/outerspacetreaty.html](http://www.unoosa.org/oosa/en/ourwork/spacelaw/treaties/outerspacetreaty.html).

find reassurance in the OST's ambiguous formulation when it comes to *private* appropriation and activities because it only bans *national* appropriation. Yet the OST is unsatisfactory for second reason: it requires states to "bear international responsibility" for private activities in space (Article VI)—an excessive government oversight which, for Rand Simberg (CEI), will "inhibit ... the development of market-based solutions."<sup>50</sup>

As shown below, the post-2000 think tanks' publications still present international space legislation as no less than a danger to the long-term survival of human civilization because of its perceived threats to property rights and sponsoring of state and interstate authority. Today's neoliberal reports on space tend to take the following steps. They first typically rhetorically construct space as central to "human flourishing" and space travel as "vital and beneficial to human well-being."<sup>51</sup> For instance, Littlewood (IEA) writes that opening the private space race would result in "enormous benefits to wealth, health and Earth's environment," while Lowe (ASI) associates it with "new scientific discovery, democratized space exploration, and much more."<sup>52</sup> Second, current international legislation is presented as thwarting efforts to unlock the promise of space by misunderstanding the beneficial influence of private property, the market, and entrepreneurship.<sup>53</sup>

Specifically, the OST and the Moon Agreement's hindrances to the instauration of strong property rights in space through their consecration of space as global commons is seen as a major obstacle to future developments of vital space activities. Like Scheraga in 1987, Peter Nelson and Walter Block use Hardin's theory of the "tragedy of the commons" (and the bison as illustrative example) to argue that space resources will be depleted and overused if no property rights are defined for them.<sup>54</sup> Property rights, they believe, create a direct interest to care for a resource and protect it in the long term. Similarly, strong property rights are thought to be the necessary condition for the development of private entrepreneurship—entrepreneurs being here again presented as the only actors (as opposed to state actors) who can develop vital space activities in an efficient way. A recurrent example is the comparison between NASA's and SpaceX's launch prices, the former being three times more expensive.<sup>55</sup> As in the 1980s, the post-2000 reports generally agree that insecure property rights will "inhibit" space entrepreneurs and private entities from pursuing the "exploration and eventual exploitation" of space.<sup>56</sup> Profits guaranteed by property rights are needed to encourage

<sup>50</sup> Simberg, "Homesteading the Final Frontier," 5. Such critique could also be found in the publications from the 1980s—see, for instance, Henson and Lucas, "Star Laws."

<sup>51</sup> Simberg, "Homesteading the Final Frontier," 4; Nelson and Block, *Space Capitalism*, 1.

<sup>52</sup> Littlewood, "True Spirit"; Lowe, "Space Invaders," 5.

<sup>53</sup> Littlewood, "True Spirit"; see also Nicolas Bouzou, "Space Is the Next Economic Revolution, and Europe Could Miss It," CAPX, Aug. 2016, at <https://capx.co/space-is-the-next-economic-revolution-and-europe-could-miss-it>; Charles N. Steele, "Space Economy Requires Property Rights, Free Market," Heartland Institute, Oct. 2019, at [www.heartland.org/news-opinion/news/space-economy-requires-property-rights-free-market](http://www.heartland.org/news-opinion/news/space-economy-requires-property-rights-free-market); Lowe, "Shoot for the Moon."

<sup>54</sup> Nelson and Block, *Space Capitalism*, 33.

<sup>55</sup> Greason and Bennett, "Economics of Space," 46.

<sup>56</sup> Simberg, "Homesteading the Final Frontier." See also Nelson and Block, *Space Capitalism*, 110; Cheng, "Outer Space and Private Property"; Lowe, "Space Invaders"; Walker-Werth, "Elon Musk Is Right." For a similar argument applied to the Moon Agreement see Dunstan, "Can We (Legally) Colonize Space?." For

the likes of Musk and Bezos to take upon themselves the enormous costs of private space ventures.

These seemingly practical critiques of the international space regime's obstructions are underpinned by a strong moral commitment to private property—especially for the contributors also identifying as libertarians. As legal scholar Jessica Whyte highlights, there is a long-running tendency amongst neoliberals to consider that property rights are an essential foundation to other human rights.<sup>57</sup> From its creation in 1947, the MPS has defined private property (along with the competitive market) as an institution without which “it is difficult to imagine a society in which freedom may be effectively preserved.”<sup>58</sup> The twenty-first-century neoliberal think-tankers follow this trend: for FEE's Walker-Werth, for instance, private property is a right “essential for human beings to be able to use their minds and live fully.”<sup>59</sup> Similarly, as shall be discussed in the second part of this article, Nelson and Block call for the establishment of property rights in space according to homesteading principles based on Rothbard's interpretation of John Locke's natural-law theory.<sup>60</sup> In placing themselves in this intellectual tradition, they designate property as a fundamental natural right, as important as life and liberty. The international space treaties thus represent a visceral threat, which explains the vehemence of their reaction.

As such, for libertarians like Nelson and Block, international space legislation should be simply axed.<sup>61</sup> The others propose more measured approaches. In typical Hayekian fashion, what is required from the state is to abandon “central planning” by dismantling the current space regime and providing instead a clear legal framework securing property rights that will foster, in Heartland Institute Charles N. Steele's words, “imagination, entrepreneurial innovation, and a free market.”<sup>62</sup>

### ***Neutralizing the OST***

The most extreme libertarians aside, and marking a slight inflection from the 1980s reports, the large majority of the post-2000 neoliberal space reports offer to preserve but “neutralize” the OST because it provides the foundation for international space law's follow-up treaties, which the US government should see as furthering its own interest—such as the Liability Convention and the Rescue Agreement.<sup>63</sup> Thus Simberg (CEI) proposes exploiting the OST's ambiguities on *national* but not *private* appropriation, by arguing that the treaty does not need renegotiation as it still allows states

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their 1980s equivalents see Bennett, “Second Space Race”; McGath, “Free Enterprise”; Copulos, “Enterprise in Orbit”; Cox, “Space Entrepreneurs.”

<sup>57</sup> Whyte, *The Morals of the Market*, 3.

<sup>58</sup> “Statement of Aims,” Mont Pèlerin Society, 8 April 1947, at [www.montpelerin.org/Statement-of-Aims.html](http://www.montpelerin.org/Statement-of-Aims.html).

<sup>59</sup> Walker-Werth, “Elon Musk Is Right.”

<sup>60</sup> For Block's tribute to Rothbard see Nelson and Block, *Space Capitalism*, xii. See also Murray Rothbard, *The Ethics of Liberty* (New York, 1998).

<sup>61</sup> Nelson and Block, *Space Capitalism*, 164.

<sup>62</sup> Steele, “Space Economy.”

<sup>63</sup> Simberg, “Homesteading the Final Frontier,” 9.

to “recognize the property claims of *anyone* who met specified conditions, regardless of their citizenship or nationality.”<sup>64</sup> In their 2017 written testimony to the US Senate, James Dunstan and Berin Szóka (TechFreedom) suggest that the United States should create an attractive domestic legal regime supporting private enterprise that other states will want to emulate, instead of directly reforming international treaties.<sup>65</sup> Greason and Bennett (Reason Foundation) support a similar approach: they advise concentrating on binational agreements between space-faring nations sharing similar interests.<sup>66</sup>

Policy-related think tanks use the aura of expertise that surrounds them to strategically achieve their political goals.<sup>67</sup> As a telling example, TechFreedom, created in 2011, specializes in providing legal expertise to influence the regulation of high-technology sectors. To defend the interests of the space industry, it employs tech lawyers and counsels like Barthold, Dunstan, and Szóka to file legal comments for requests of information and comment issued by governmental agencies (e.g., NASA, the Federal Communications Commission (FCC), or the White House’s Office of Science and Technology Policy) and amicus briefs in courts, as well as to provide testimonies as legal experts in front of US Congress’s specialized committees or to draft letters to members of Congress proposing technical bill amendments.<sup>68</sup>

The lobbying efforts of the US neoliberal think tanks and their New Space corporate allies have started to bear fruit. In 2015, Congress adopted the Commercial Space Launch Competitiveness Act (CSLCA), celebrated as a “victory for commercial space” by TechFreedom.<sup>69</sup> The CSLCA “promote[s] the rights of United States citizens to engage in commercial exploration for and commercial recovery of space resources.”<sup>70</sup> It thus opens the way to recognizing the appropriation of space resources

<sup>64</sup>Ibid., 10, original emphasis.

<sup>65</sup>Dunstan and Szóka, “Written Testimony.”

<sup>66</sup>Greason and Bennett, “Economics of Space,” 76.

<sup>67</sup>Plehwe, “Think Tank Networks and the Knowledge–Interest Nexus.”

<sup>68</sup>For legal comments see “NASA Lacks Authority to Independently Develop Non-interference Standards for Lunar Operations,” TechFreedom, 7 June 2024, at <https://techfreedom.org/nasa-lacks-authority-to-independently-develop-non-interference-standards-for-lunar-operations>; “FCC Lacks Statutory Authority and Expertise for Outer Space Activities,” TechFreedom, 30 April 2024, at <https://techfreedom.org/fcc-lacks-statutory-authority-and-expertise-for-outer-space-activities>. For 2021 and 2023 amicus briefs in support of SpaceX see Barthold, Dunstan, and Szóka, “Brief of Amicus Curiae IBFS File Nos. SAT-LOA-20200526-00055”; Corbin Barthold, James E. Dunstan, and Berin Szóka, “Brief of Amicus Curiae Techfreedom in Support of Appellee/Respondent and Affirmance on Appeal from the Federal Communications Commission, IBFS File No. SAT-MOD-20200417-00037,” 28 Sept. 2021, at <https://techfreedom.org/wp-content/uploads/2021/09/File-Stamped-TechFreedom-Amicus-Brief-Viasat-v-FCC.pdf>. For legal testimonies to Congress see Dunstan and Szoka, “Written Testimony”; James E. Dunstan, “Testimony of TechFreedom James E. Dunstan before the House Committee on Space, Science, and Technology,” 13 July 2023, at <https://techfreedom.org/wp-content/uploads/2023/07/Space-Governance-Testimony-July-13-2023.pdf>. For amendment proposal see Berin Szóka and James E. Dunstan, “Letter Re Amendments to House Representatives,” 19 May 2015, at [http://docs.techfreedom.org/TF\\_Letter\\_Re\\_Amendments\\_to\\_HR\\_1508.pdf](http://docs.techfreedom.org/TF_Letter_Re_Amendments_to_HR_1508.pdf).

<sup>69</sup>“Space Property Legislation a Victory for Commercial Space, but Has Troubling Loophole,” TechFreedom, 11 Nov. 2015, at <https://techfreedom.org/space-property-legislation-a-victory-for>.

<sup>70</sup>“U.S. Commercial Space Launch Competitiveness Act,” U.S. Congress, Nov. 2015, at [www.congress.gov/114/plaws/publ90/PLAW-114publ90.pdf](http://www.congress.gov/114/plaws/publ90/PLAW-114publ90.pdf).

by individuals and commercial entities. It also circumvents the constraints set by the OST by stipulating that the US will not thereby assert sovereignty or jurisdiction over any celestial body. In April 2020, President Trump followed up with an executive order stipulating that the US “does not view [outer space] as a global common” and will “encourage international support for the public and private recovery and use of resources in outer space.”<sup>71</sup> Bandow greeted this clarification as a “good step forward.”<sup>72</sup> Finally, in October 2020, the US, along with seven other founding members (including the United Kingdom and Luxembourg) signed the Artemis Accords, which endorse the strategy of bilateral agreements that TechFreedom and the Reason Foundation advocate. Under the objective of establishing a framework for cooperation in the civil exploration and peaceful use of the Moon, Mars, and other astronomical bodies, the accords sanction the US reinterpretation of the OST.<sup>73</sup> Specifically, it guarantees a right to extract and appropriate abiotic resources like minerals and water under the reasoning that extraction “does not inherently constitute national appropriation under Article II of the Outer Space Treaty”—an argument that Szóka and Dunstan have also made, as in their 2015 letter to the US House representatives.<sup>74</sup> Similarly, it creates “safety zones” of noninterference and exclusion around areas of (for instance, mining) operations, here again mirroring suggestions that neoliberal space consultants have made.<sup>75</sup> Crucially, the US has conditioned states’ future collaborations on NASA-led projects, such as the Artemis Program for lunar exploration, to their compliance with the principles set forth in the Artemis Accords.<sup>76</sup> At the time of publication, sixty-eight states have signed, reinforcing the US push for the introduction of private property in space by making it international customary law.<sup>77</sup>

### Appropriating the final frontier: space as a fantasy

The second and third part of this article adopt a more thematic approach, to determine why space keeps being an object of fascination for twenty-first-century think-tankers. A first broad answer is that space has a special imaginative and affective texture. It acts as an empty screen onto which fantasies of myriad futures can be projected—something the neoliberal think-tankers tap into when focalizing their lobbying efforts onto the international space regime. Yet, as I show, the utopian speculations they associate with space as the “final frontier” are riddled with intellectual anxiety.

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<sup>71</sup>“Executive Order on Encouraging International Support for the Recovery and Use of Space Resources,” White House, April 2020, at <https://trumpwhitehouse.archives.gov/presidential-actions/executive-order-encouraging-international-support-recovery-use-space-resources>.

<sup>72</sup>Bandow, “Trump EO.”

<sup>73</sup>Boley and Byers, “U.S. Policy.”

<sup>74</sup>“Artemis Accords,” NASA, 13 Oct. 2020, at [www.nasa.gov/wp-content/uploads/2022/11/Artemis-Accords-signed-13Oct2020.pdf?emrc=69dfa8ab50a8b](http://www.nasa.gov/wp-content/uploads/2022/11/Artemis-Accords-signed-13Oct2020.pdf?emrc=69dfa8ab50a8b); Szóka and Dunstan, “Letter.”

<sup>75</sup>NASA, “Artemis Accords”; Szóka and Dunstan, “Letter.”

<sup>76</sup>Rossana Deplano, “The Artemis Accords: Evolution or Revolution in International Space Law?” *International and Comparative Law Quarterly* 70/3 (2021), 799–819.

<sup>77</sup>“Artemis Accords,” *nasa.gov*, at [www.nasa.gov/artemis-accords](http://www.nasa.gov/artemis-accords) (accessed 7 July 2026).



### *Space as the final frontier*

The frontier is a widespread trope in the neoliberal think-tank literature on space.<sup>78</sup> It participates in what political economist Jamie Peck calls the “contemporary cultural and political recoding of that American frontier spirit, one that has long valorized movement *on and out*.”<sup>79</sup> While the motif of the frontier is not specific to neoliberal thought, its recurrence in neoliberal works on space calls for further analysis. Cato’s 2002 *Space: The Free-Market Frontier*, the CEI’s 2012 *Homesteading the Final Frontier* and Greason and Bennett’s 2019 Reason Foundation report all explicitly connect space with the American “frontier” and unambiguously celebrate it.<sup>80</sup> Nelson and Block recall how, “From the seventeenth through the nineteenth centuries, based on the private settlements on the frontier in North America, a new commitment to freedom spread worldwide.”<sup>81</sup> Similarly, Greason and Bennett praise the frontier of the “American West” and its “pioneers,” along with the “sea” frontier traversed by European navigators of the fifteenth and sixteenth centuries, for catalyzing economic development.<sup>82</sup> These are the models that outer space development should follow. I distinguish two different dimensions in the neoliberal conceptualizations of space as a “frontier,” as a geographical concept and as an affective mobilizer.

In the explicitly libertarian circles, the motif of the American frontier is used to conceptualize the boundary between corrupt failing states on the one hand and new territories where a utopia might be created on the other. Walker-Werth (FEE), for instance, depicts how, “centuries ago,” there were still “areas of Earth’s surface that were not ruled by governments—areas beyond laws, regulations, taxes, kings, and politicians. Those frontiers provided an opportunity to set up new societies based on radical ideas, free from the trappings of the old world.”<sup>83</sup> The US is, for Walter-Werth, the “shining example” of such places. To draw on Matthew Johnson’s work, outer space is assimilated to these extralegal spaces of exception, placed beyond the state.<sup>84</sup> These spaces function as a blank sheet on which like-minded “freedom-loving” communities can escape the smothering grip of the state and its normative rules to redefine the values according to which they want to live.<sup>85</sup>

The rules of the free market will order the space communities imagined by neoliberal and libertarian authors. As Alexander Salter and Peter Leeson argue in *Cato Journal*, the absence of sovereign states in the “celestial anarchy” that they champion

<sup>78</sup>The frontier is a trope within the larger New Space lobby. See Johnson, “Mining the High Frontier,” 182–211; Ben Little and Alison Winch, *The New Patriarchs of Digital Capitalism: Celebrity Tech Founders and Networks of Power* (Abingdon, 2021), 144–66; Alina Utrata, “Engineering Territory.”

<sup>79</sup>Jamie Peck, “Neoliberal Suburbanism: Frontier Space,” *Urban Geography* 32/6 (2011), 884–919, at 909, original emphasis. See also Quinn Slobodian, *Crack-Up Capitalism: Market Radicals and the Dream of a World without Democracy* (Dublin, 2023).

<sup>80</sup>Hudgins, *Space*; Simberg, “Homesteading the Final Frontier”; Greason and Bennett, “Economics of Space.”

<sup>81</sup>Nelson and Block, *Space Capitalism*, 21.

<sup>82</sup>Greason and Bennett, “Economics of Space,” 14–16.

<sup>83</sup>Walker-Werth, “Elon Musk Is Right.”

<sup>84</sup>Johnson, “Mining the High Frontier,” 182–211.

<sup>85</sup>Nelson and Block, *Space Capitalism*, 21.

will not result in lawlessness. Private arbitration associations such as the International Chamber of Commerce, whose operations transcend states' legal boundaries, prove that the private sector can organize itself to guarantee property rights.<sup>86</sup> Private arbitration thus is an effective alternative to state rule. As another example of independent space communities, Walker-Werth and Barthold (TechFreedom) find Musk's self-governing Martian settlement promising. As the owner of the settlement's infrastructure, Musk would also be able to determine its rules, thus acting as its *de facto* sovereign.<sup>87</sup> Walker-Werth and Barthold seem unfazed by the risks of Musk overreaching. His status as a star entrepreneur seems enough for them to believe he will uphold the free market as a fundamental governing principle. These space reports follow a larger trend in neoliberal libertarian circles—recently analyzed by historian Quinn Slobodian—characterized by its championing of new sovereign communities seceding from states and organized around the model of the corporation, where voting power is concentrated in the hands of shareholders.<sup>88</sup> Contrasting—in Nelson and Block's case explicitly—their ideal space community with the model of the commons (in which no ownership is deemed to lead to an overuse or waste of resources), private ownership becomes the sign of having a stake in the long-term future of the community and thus comes with a sovereign right to shape it.<sup>89</sup>

A striking sense of settler colonial nostalgia also transpires from the libertarian prose. As Walker-Werth deplors, "There are no new frontiers left in which to establish a freer kind of society" on Earth today.<sup>90</sup> Or, in the words of fellow libertarian Peter Thiel (close to the neoliberal thought collective), "there are no truly free places left in our world."<sup>91</sup> All territories have been claimed by states and submitted to their sovereign rule. For Thiel, "the mode for escape [from world politics] must involve some sort of new and hitherto untried [technological] process that leads us to some undiscovered country."<sup>92</sup> Outer space plays such a role as its "vast reaches ... represent a limitless frontier ... [as well as] a limitless possibility for escape."<sup>93</sup> For Walker-Werth, space is the final frontier that makes up for the disappearance of frontiers on Earth.<sup>94</sup>

<sup>86</sup>Salter and Leeson, "Celestial Anarchy."

<sup>87</sup>Alina Utrata, "Company Rules: Martian Colonies and Silicon Valley's Corporations in Space," *PERCblog*, 2022, at [www.goldperc.uk/project\\_posts/company-rules-martian-colonies-and-silicon-valleys-corporations-in-space](http://www.goldperc.uk/project_posts/company-rules-martian-colonies-and-silicon-valleys-corporations-in-space).

<sup>88</sup>Slobodian, *Crack-Up*.

<sup>89</sup>Nelson and Block, *Space Capitalism*, 48 n. 23.

<sup>90</sup>Walker-Werth, "Elon Musk Is Right."

<sup>91</sup>Peter Thiel, "The Education of a Libertarian," *Cato Unbound*, April 2009, at [www.cato-unbound.org/2009/04/13/peter-thiel/education-libertarian](http://www.cato-unbound.org/2009/04/13/peter-thiel/education-libertarian). Peter Thiel was awarded the 2015 Hayek Lifetime Achievement Award by the Austrian Economic Center and participated in the 2018 and 2020 meetings of the Mont Pèlerin Society. See "Annual Gala: Peter Thiel Awarded by the Austrian Economics Center," Austrian Economic Center, Nov. 2015, at [www.austriancenter.com/annual-gala-peter-thiel-awarded-by-the-austrian-economics-center](http://www.austriancenter.com/annual-gala-peter-thiel-awarded-by-the-austrian-economics-center); Peter Thiel and Peter Robinson, "China, Globalization, Capitalism, Silicon Valley, Political Correctness, and Exceptionalism," in *The Mont Pèlerin Society: From the Past to the Future* (Stanford, 2020), 534–45.

<sup>92</sup>Thiel, "The Education of a Libertarian." See also Harrison Smith and Roger Burrows, "Software, Sovereignty and the Post-neoliberal Politics of Exit," *Theory, Culture and Society* 38/6 (2021), 143–66.

<sup>93</sup>Thiel, "The Education of a Libertarian."

<sup>94</sup>Walker-Werth, "Elon Musk Is Right."

Beyond the libertarian circles, the motif of the frontier is used in neoliberal space reports to trigger certain affects and mobilize their American readers. The myth of the frontier appeals to a sense of collective pride in what authors and readers perceive as the glorious achievements of Manifest Destiny.<sup>95</sup> As literary theorist Roland Barthes argues, myths have a hailing function: operating on an affective and emotional level, they include us, spectators, in an indistinctive and uncritical universality.<sup>96</sup> Specifically, the neoliberal think-tankers tap into the epics of adventure allegedly associated with the frontier.<sup>97</sup> For instance, Christopher Talgo (Heartland Institute) praises “America” for “embrac[ing] exploration and ingenuity” since “its founding”; he recounts the “European settlers [who] ventured to the New World despite the hardships that awaited them,” who “braved the elements and many other threats.”<sup>98</sup> Space privatization would allow the US to “continue to push the boundaries of discovery.”<sup>99</sup> The figure of the space entrepreneur (more so than the astronaut) is thus uncritically lionized as the one thanks to whom the limit between the known and the unknown will be overcome, leading to developments in the field of health, technologies, and sciences, and providing access to new natural resources.

Yet, as Dunstan makes clear, science actually comes second: the commercial exploitation of space resources should remain the primary objective of space exploration.<sup>100</sup> For Simberg, space should not be conceptualized as a “scientific preserve,” but rather as a “new venue for human expansion.”<sup>101</sup> His remark illustrates well what Marxist theorist Jason Moore describes as capitalism’s dependency on “frontiers of cheap nature.”<sup>102</sup> Because of its reliance on fantasies of endless accumulation and consumption, capitalism expands outwards both to find raw material, energy, and labor, and to create new markets, a tendency that Marxist geographer David Harvey conceptualizes as the “spatial fix.”<sup>103</sup> Sociologists Peter Dickens and James Ormrod use this idea in their work to think about the function of space in the capitalist imaginary.<sup>104</sup> Like the other frontiers celebrated in neoliberal thought, space as the “limitless frontier” must thus be read as a cultural project that justifies and glorifies appropriation.<sup>105</sup> According to this anthropocentric fantasy, the universe is amenable to entrepreneurial agency; outer space can be reached, understood, and, ultimately, controlled.

Of course, both geographical and affective evocations of the frontier erase the most problematic historical dimensions of the age of “discovery” and settlement. This is

<sup>95</sup>See, for instance, Rubenstein, *Astrotopia*, 69–94; Greg Grandin, *The End of the Myth: From the Frontier to the Wall in the Mind of America* (New York, 2019).

<sup>96</sup>Roland Barthes, *Mythologies* (Paris, 2007), 211–72.

<sup>97</sup>See Johnson, “Mining the High Frontier,” 184.

<sup>98</sup>Talgo, “A New Era.”

<sup>99</sup>*Ibid.* See also Lilico, “Is Colonising Space Economically Viable?”

<sup>100</sup>Dunstan, “Comments.”

<sup>101</sup>Simberg, “Homesteading the Final Frontier,” 4.

<sup>102</sup>Jason W. Moore, “World Accumulation and Planetary Life, or, Why Capitalism Will Not Survive until the ‘Last Tree Is Cut,’” *IPPR Progressive Review* 24/3 (2017), 176–202, at 179.

<sup>103</sup>*Ibid.*, 178. David Harvey, “The Spatial Fix: Hegel, von Thünen, and Marx,” *Antipode*, 13/3 (1981), 1–12.

<sup>104</sup>Peter Dickens and James S. Ormrod, *Cosmic Society: Towards a Sociology of the Universe* (London, 2007).

<sup>105</sup>Moore, “World Accumulation,” 182.

particularly obvious in their uncritical and dehistoricized celebration of space “colonization.”<sup>106</sup> The word “colonization” and its derivatives are commonly used in these reports, often without a second thought given to the horrors of the colonial past. For instance, Lilico limits his column on the colonization of Mars to the question of economic viability.<sup>107</sup> When seemingly addressing the issue, Nelson and Block make a historically unjustifiable distinction between pernicious state-led colonization—the colonization of the “Spanish and Portuguese conquistadores [which] despoiled native lands, killed inhabitants, and stole their valuables”—and benign private colonization by “freedom-loving” English settlers.<sup>108</sup> Even when conceding that “[c]olonization” has a bad press” because of its association “with exploiting helpless people” (although they swiftly reference MPS-member Niall Ferguson’s “argument that this process was actually helpful to its supposed victims”), they argue that it happily will not have such deleterious effect in outer space because there are no natives—no “Martians or Venusians”—to dispossess.<sup>109</sup> As Matthew Johnson writes in his study of the New Space lobby, these fantasies of “guilt-free appropriation” are related to the depoliticizing process of “mythologization” denounced by Barthes.<sup>110</sup> For Barthes, myths do not hide colonialism itself, only its “contingent, historical, or in one word *fabricated*, quality”; their function is not to “deny things” but, on the contrary, “to purify, exculpate and found them in nature and eternity.”<sup>111</sup>

### **Property rights reloaded**

Yet the neoliberal appropriation of outer space might not be as devoid of “guilt” as its proponents delude themselves. The myth’s purification process leaves a remainder that cannot be fully repressed. The inherent violence of appropriation remains a haunting presence in the narratives analyzed. Yes, there are no natives to dispossess, but because of space’s peculiar legal status as *res communis* it is humanity that would be the victim of extra-planetary robbery. The appropriation of celestial bodies particularly runs counter to other traditional uses and conceptualizations of space. The two examples that are often mentioned in the academic literature are the way many Indigenous cultures conceptualize celestial bodies as spiritual beings or as relatives, and the way the light pollution created by the multiplication of satellites in the sky, such as Musk’s Starlink constellation, endangers traditional celestial navigation.<sup>112</sup>

As private property is such an important pillar of neoliberal thought, any doubt or concern about the fundamental arbitrariness and violence of initial appropriation may constitute a genuine epistemological and normative threat for the neoliberal theoretical

<sup>106</sup>See also Utrata, “Engineering Territory.”

<sup>107</sup>Lilico, “Is Colonising Space Economically Viable?”

<sup>108</sup>Nelson and Block, *Space Capitalism*, 21–2. They mention colonization and its derivative no less than seventy times in their book. For other analyses of the neoliberal differentiation between good and bad colonization see Lars Cornelissen, “Fossil Neoliberalism: Towards an Intellectual History of Neoliberal Climate Change Denial,” this issue; Cornelissen, *Neoliberalism and Race* (Stanford, 2025); and Slobodian, *Globalists*.

<sup>109</sup>Nelson and Block, *Space Capitalism*, 142 n. 14.

<sup>110</sup>Johnson, “Mining the High Frontier,” 209–10.

<sup>111</sup>Barthes, *Mythologies*, 253, my translation, original emphasis.

<sup>112</sup>Utrata, “Engineering Territory,” 7; Janssen and Yap, “Governing Outer Space,” 34.

project. This is what is at play in the neoliberal and libertarian debates on homesteading in space—which amply build on hailing spatial fantasies, like the myth of the frontier. These texts go to great lengths to prove the morality of private appropriation of space resources. “Homesteading” as a concept is itself closely associated with the history of white settlements and the dispossession of Indigenous people on the American frontier, as for instance represented by the US 1841 Preemption Act and 1862 Homestead Act, which gave settlers the right to appropriate frontier lands as extralegal spaces, providing settlers continuously occupied and “improved” these lands.<sup>113</sup> Continuing the ideals expounded in Locke’s *Two Treatises of Government*, they established that “fructifying” the land through cultivation or extraction dispenses a moral right (and therefore property rights) to territory, especially if it is unoccupied or “wasted.”<sup>114</sup> In Locke’s vivid description, as owner of himself and of his labor, the settler who works the land has “mixed his labour with and joined to it something that is his own, and thereby makes it his property.”<sup>115</sup> Locke is an explicit reference in several post-2000 neoliberal space reports—as in the two variations on homesteading examined below.

As previously discussed, self-described libertarians such as Nelson, Block, and Walker-Werth adopt Murray Rothbard’s Lockean approach to private property and homesteading.<sup>116</sup> Reusing Locke’s metaphor of mixing one’s labor with the land, they defend appropriation on an uncomplicated and unapologetic “first in use, first in right” basis.<sup>117</sup> The first settlers would also be granted the power, with a long-term impact, of deciding how to use the land they control—for instance, if they want to use it for mining, terraforming for settlement or tourism, or preserving as an environmental treasure.<sup>118</sup> Their plan would therefore provide an exclusive legal right to appropriate a resource-rich environment currently held in common to a small minority of stratospherically rich individuals, like Bezos or Musk, who are the only private actors who can currently afford partaking in space ventures. This sci-fi reenactment of primitive accumulation via colonial enclosure would certainly deepen the stark inequalities already existing in today’s world. Yet Walker-Werth, and Nelson and Block, still find the need to rhetorically circumvent such accusation: appropriation would be justified as these first settlers-cum-terraformers would necessarily, through their laborious efforts, improve a barren and infertile land, thus serving the greatest number in the

<sup>113</sup>Robert Nichols, “Theft Is Property! The Recursive Logic of Dispossession,” *Political Theory* 46/1 (2018), 3–28.

<sup>114</sup>*Ibid.*, 19. For studies on how Locke’s work was used to justify colonial dispossession in the English-speaking world see David Armitage, “John Locke Theorist of Empire?,” in Sankar Muthu, ed., *Empire and Modern Political Thought* (Cambridge, 2012), 84–111; and Barbara Arneil, “John Locke, Natural Law and Colonialism,” *History of Political Thought* 13/4 (1992), 587–603.

<sup>115</sup>John Locke, *Two Treatises on Government and A Letter Concerning Toleration*, ed. Ian Shapiro (New Haven, 2003), 111–12.

<sup>116</sup>See Rothbard, *The Ethics of Liberty*. Rothbard is a recurring reference in Nelson and Block’s *Space Capitalism*.

<sup>117</sup>Nelson and Block, *Space Capitalism*, 6, 29.

<sup>118</sup>Walter Block and Jacon H. Huebert, “Environmentalists in Outer Space,” Foundation for Economic Education, March 2008, at <https://fee.org/articles/environmentalists-in-outer-space>. This decision would also impact the way surrounding land is used. For instance, an environmentalist who homesteads next to an already established mine cannot complain about preexisting neighboring activities.

long term.<sup>119</sup> Eventually, how much land they can directly transform to “make it useful” would be a natural limit to how much they can appropriate.<sup>120</sup>

Lowe offers a more subtle understanding of appropriation, but her proposal for distributing Moon lands’ property rights is haunted by the specter of colonial dispossession: “debate rages about the property-related injustices of the past, here on Earth. Debates about the actions of our forebears, relating to the acquisition of property, the distribution of access to resources, and the colonisation of areas already serving as the livelihood and homes of indigenous peoples.”<sup>121</sup> Fully aware of “the costs that property rights regimes can impose,” she wants to find ways to ensure that “legal claims to ownership are also morally justified.” Space represents such an opportunity. As a blank page devoid of previous inhabitations, institutions, and history, it gives us “a one-time chance to get matters of property right, this time, from the off.” In other words, space has the potential to save private property from its shameful past. She proposes to avoid the limitations of the “first come, first served” principle by creating a competitive and market-driven system of rent in which an international committee would allocate lunar territory on a temporary basis to entrepreneurs who, as in Locke, shall prove that they enhance the land.<sup>122</sup> The rents would go to a common fund that would help currently less privileged individuals to also compete to become lunar tenants in the future.

In both variations, we can see that the speculative future uses of space are evoked to address the unstable core of neoliberal thought. Concerns about the sanctity of private property relating to its tarnished past are remedied by phantasmatic visions of justly developing the bounty of far-off worlds. Space acts as a fantasy—that is, a psychic mechanism through which gaps that mark the inconsistencies of a subject’s representations are contained, thereby providing a sense of coherence and wholeness—and therefore allows neoliberals to discard issues that could unsettle some basic tenets of their worldview.<sup>123</sup>

### “That was a no-brainer”: using space as an environmental fix

Climate change is another thorny issue for neoliberal hegemony that the fantasy of space can resolve. It is no coincidence that many of the think tanks mentioned, like the Heartland Institute or the Competitive Enterprise Institute, are well-known centers of climate denial.<sup>124</sup> For instance, Heartland’s Charles Steele co-authored the 2019 report of the climate-change-sceptic Nongovernmental International Panel on Climate

<sup>119</sup> Nelson and Block, *Space Capitalism*, 147–8; Walker-Werth, “Elon Musk Is Right.”

<sup>120</sup> Walker-Werth, “Elon Musk Is Right.”

<sup>121</sup> Lowe, “Shoot for the Moon.”

<sup>122</sup> Lowe, “Space Invaders.”

<sup>123</sup> I draw here on the Lacanian tradition of ideology critique. See, for recent examples, Erik Swyngedouw, “The Unbearable Lightness of Climate Populism,” *Environmental Politics* 31/5 (2022), 904–25; Robert Watt, “The Fantasy of Carbon Offsetting,” *Environmental Politics* 30/7 (2021), 1069–88.

<sup>124</sup> Naomi Klein, *This Changes Everything: Capitalism vs. the Climate* (London, 2014); Andreas Malm and Zetkin Collective, *White Skin, Black Fuel: On the Danger of Fossil Fascism* (London, 2021).

Change (NIPCC), *Climate Change Reconsidered II*.<sup>125</sup> Elsewhere, he asserts that fossil fuels are “on net, positive, and make humans better off,” while “current proposals for addressing climate change impose costs on people that far outweigh benefits.”<sup>126</sup>

As in the case of the OST, neoliberal think-tankers tend to see any state-led coordinated action to tackle climate change as a potential threat to individual freedom and the efficient workings of the free market. Steele targets the 2019 report of the International Panel on Climate Change (IPCC)—especially its call to scale down the use of fossil fuels through the funding of alternative energy sources, taxation, and social justice—as “a call for global central planning and income redistribution—a sort of ‘socialism lite’ dressed up as sustainable development.”<sup>127</sup> The IPCC recommendations endanger economic freedom and the free market, and, through them, “human prosperity around the world.”

To defuse the threat, neoliberal think-tankers strategically present carbon-fueled private entrepreneurship in space as a more efficient way to address the issue of climate change than regulations to curb carbon emissions. As often with neoliberal advocates, spreading market mechanisms to new domains is presented as the only adequate solution to the climate crisis.<sup>128</sup> Here again, marketized space enables capitalism to bypass planetary limits to growth.<sup>129</sup> Below I explore ways the neoliberal commentators use space to circumvent the problem of climate change, at least on a phantasmatic level.

### *Solving the limitation of earthly resources*

Most of the texts analyzed advertise the appropriation of space land primarily for its mining potential. Asteroids could provide access to “rare earth material such as neodymium, scandium, and yttrium,” as well “platinum-group metals such as platinum itself and palladium.”<sup>130</sup> The Moon could also be used as a convenient low-gravity extraterrestrial refill station for aluminum, titanium, nickel, and iron, as well as water, hydrogen, and oxygen needed for longer outer-space journeys and infrastructure.<sup>131</sup> The think-tankers are quick to emphasize how lucrative space mining could be. Littlewood speculates that “a single, fully mined asteroid would be worth more than \$800 billion,” while Nelson and Block reckon that some asteroids could “contain \$20

<sup>125</sup> Roger Bezdek, Craig D. Idso, David R. Legates, and S. Fred Singer, *Climate Change Reconsidered II: Fossil Fuels* (Arlington Heights, 2019). For an analysis of the NIPCC see Plehwe, “Think Tank Networks and the Knowledge–Interest Nexus.”

<sup>126</sup> Charles N. Steele, “Climate Doom Ahead? Think Twice,” Heartland Institute, 17 Jan. 2019, at [www.heartland.org/news-opinion/news/climate-doom-ahead-think-twice?source=policybot](http://www.heartland.org/news-opinion/news/climate-doom-ahead-think-twice?source=policybot).

<sup>127</sup> Ibid.

<sup>128</sup> Philip Mirowski, *Never Let a Serious Crisis Go to Waste: How Neoliberalism Survived the Financial Meltdown* (London, 2013), 334–42.

<sup>129</sup> James S. Ormrod, “Outer Space and New Frontiers to Environmental Imaginations,” in Katharine Legun, Julie C. Keller, Michael Carolan, and Michael M. Bell, eds., *Cambridge Handbook of Environmental Sociology* (Cambridge, 2020), 243–61.

<sup>130</sup> Simberg, “Homesteading the Final Frontier,” 7. See also Nelson and Block, *Space Capitalism*, 102–5; Greason and Bennett, “Economics of Space,” 6–7; Littlewood, “True Spirit.”

<sup>131</sup> Greason and Bennett, “Economics of Space,” 8–9; Nelson and Block, *Space Capitalism*, 67.



trillion worth of valuable metals.”<sup>132</sup> Simberg, and Greason and Bennett, also highlight the strategic dimension for the US of investing in space mining to source rare-earth materials (which are used to manufacture lasers, chips, and batteries) at a time when their supply on Earth is dominated by China.<sup>133</sup>

What makes these materials so precious is their relative scarcity on Earth. As such, the reports tend to connect their promotion of space mining with reflections on the limitation of earthly resources, but with an important twist that reflects the influence of business professor and MPS member Julian Simon on neoliberal environmental thought.<sup>134</sup> Simon is well known for his cornucopian theory on the infinitude of natural resources, which he expounded in *The Ultimate Resource* (1981). The book was constructed as a challenge to neo-Malthusian theories on resource scarcity and overpopulation as found in the works of Meadows’s *Limit to Growth* (1972) or Ehrlich’s *Population Bomb* (1968).<sup>135</sup> For Simon, it is not “economically meaningful” to conceptualize natural resources as quantitatively limited as we humans “have the capacity to develop additional ways to meet our needs” through technological innovation.<sup>136</sup> While some resources might have appeared physically limited at given moments in time, these impressions, along with the conceptualizations and calculations of what this “limit” consisted in, were always proven wrong.<sup>137</sup> If a resource that seems scarce is truly needed, it will be given a monetary value that will encourage enterprising individuals to find ways to produce it by other means (e.g. synthetic production or recycling processes), or find an alternative to it (for instance by discovering or creating new materials).<sup>138</sup>

Importantly, outer space is already at the center of Simon’s provocative argument. As he wrote in 1981, in the unlikely event that specific natural resources on Earth become exhausted, “there is the possibility that humans will come to exploit the resources of other parts of the cosmos, which is so huge relative to the solar system as to render calculations irrelevant under any conceivable rate of growth.”<sup>139</sup> Space is here clearly used as a trump card that renders discussions on resource limitation “irrelevant.” Even if Simon were to be proven wrong and a resource could not be sourced by any alternative on Earth, we could expand “the boundaries of the system from which we derive resources” in the direction of “the Moon.”<sup>140</sup> Space creates the potential for infinite accumulation.

Today’s neoliberal think-tankers enthusiastically take up Simon’s argument—and not always in the most subtle way. For instance, Nelson and Block adopt an extreme-Simonian perspective when, citing Simon’s work, they argue that “running out of

<sup>132</sup> Littlewood, “True Spirit”; Nelson and Block, *Space Capitalism*, 109.

<sup>133</sup> Simberg, “Homesteading the Final Frontier,” 7; Greason and Bennett, “Economics of Space,” 6–7.

<sup>134</sup> For evidence of Simon’s membership in the MPS see *Newsletter of the Mont Pèlerin Society* 1 (1991), 2.

<sup>135</sup> Troy Vetteze, “Hayek against Malthus: Julian Simon’s Neoliberal Critique of Environmentalism,” *Critical Historical Studies* 10/2 (2023), 283–311.

<sup>136</sup> Julian Simon, *The Ultimate Resource* 2 (Princeton, 1996), 52.

<sup>137</sup> *Ibid.*, 66.

<sup>138</sup> *Ibid.*, 63.

<sup>139</sup> *Ibid.*, 65.

<sup>140</sup> *Ibid.*, 66.

resources' here, back on the home planet, is a fallacious notion" as "[w]e are at present (2016) up to our armpits in resources, and hardly need to venture out into the heavenly bodies to obtain any more"; they explain that their motivation to colonize space is by contrast purely ideological.<sup>141</sup> More to the point, in 2018, Gale L. Pooley and Marian L. Tupy from the Cato Institute launched a "Simon Abundance Index" to measure "the change in abundance of resources over a period of time."<sup>142</sup> Unsurprisingly, they reach the very Simonian conclusion that, despite rising climate concerns, "the Earth was 379.6 percent more abundant in 2017 than it was in 1980." Like Simon, they acknowledge that "[t]he Earth is a closed system," but that "[o]ne day, we might be able to replenish our resources from outer space by, for example, dragging a mineral-rich asteroid down to Earth."

James Dunstan also follows Simon's steps when linking space with the infinitude of its natural resources. For example, he compares space with a "pie [that] is nearly infinite."<sup>143</sup> According to him, "Opening the space frontier brings access to the *near-limitless* energy of the sun, and the ability to harvest the quintillions worth of rare Earth metals contained in asteroids that will fuel the future hydrogen and fusion economies of Earth."<sup>144</sup> Crucially, Dunstan makes a direct connection between the commercialization of space and future environmental policies—a connection that is also found in Simon's work. When writing in 1981, Simon explicitly challenged Ehrlich's and Meadows's concerns by provocatively stating that "the [1981] world [was] creating new resources and cleaning up the environment at an ever-increasing rate."<sup>145</sup> In other words, he argued that individuals and communities could always "overcome environmental problems" by finding new resources and technologies.<sup>146</sup> Dunstan also links entrepreneurial activity in space with the development of carbon-free ("hydrogen and fusion") industries on Earth. He adds, "Such advances [in space] will allow us to wean ourselves off ugly fossil fuels without the economic devastation forecast if we use only terrestrial renewables," for instance by developing solar power capture beyond the Earth's surface.<sup>147</sup>

### ***Extraterrestrial dumpsters***

Another way in which neoliberals envision space to solve Earth's most pressing socio-environmental problems is by offering extraterrestrial uninhabited territories towards which some of these issues might conveniently be evacuated. As Greason and Bennett

<sup>141</sup> Nelson and Block, *Space Capitalism*, 223.

<sup>142</sup> Gale L. Pooley and Marian L. Tupy, "The Simon Abundance Index: A New Way to Measure Availability of Resources," Cato Institute, Dec. 2018, at [www.cato.org/policy-analysis/simon-abundance-index-new-way-measure-availability-resources](http://www.cato.org/policy-analysis/simon-abundance-index-new-way-measure-availability-resources).

<sup>143</sup> Dunstan, "Can We (Legally) Colonize Space?". He adds that "there's a trillion times more resources in the near solar system than there is on earth."

<sup>144</sup> James E. Dunstan, "Bring on the Space Barons," *TechFreedom on Medium*, Sept. 2021, at <https://medium.com/@TechFreedom/bring-on-the-space-barons-e425129fbff6>, emphasis added.

<sup>145</sup> Simon, *The Ultimate Resource*, 16.

<sup>146</sup> *Ibid.*, 16.

<sup>147</sup> Dunstan, "Space Barons." For more details on solar-power satellites see also Greason and Bennett, "Economics of Space."

write, “global solutions to Earth’s challenges, especially environmental ones, are highly constrained when limited to the planet’s surface, and must be continually weighed against human welfare—especially in developing nations. Using the entire solar system seems in general more apt to result in a cleaner and safer Earth environment.” They use mining as an example. Whereas mining resources on Earth has clear “environmental impact externalities” (which led to temporary closures of US earth elements mines in the 2000s and 2010s), mining space resources instead can almost be presented as an act of social justice as it would transfer polluting mining activities into “airless, lifeless places.”<sup>148</sup>

Others, like Dunstan, mention how the environmental cost of alternatives to fossil fuels, such as nuclear waste generated by “fusion,” could be solved through the marketization of space.<sup>149</sup> Block and Huebert (for the FEE), for instance, propose that we send nuclear waste to Venus (aboard private rockets, of course!)—a planet whose atmosphere is already toxic to humans, and which could therefore not be more polluted than it already is.<sup>150</sup> The risk of nuclear convoys exploding in Earth’s atmosphere is dismissed as the “risk would shrink as the private sector moves further into space transportation.”<sup>151</sup> They also recommend moving the most toxic and polluting forms of manufacturing in space, an idea that Dunstan, Littlewood, and Greason and Bennett also embrace.<sup>152</sup> The latter also suggest moving dangerous scientific activities, such as manipulating viruses and advanced fusion, off Earth to create a protective “vacuum gap.”<sup>153</sup>

Once again, the extralegal and extra-human status of space turns it into an imaginary speculative terrain in which today’s problems will magically be solved tomorrow.

### ***Speculative futures***

Think-tankers’ reports are characterized by their optimistic faith in the power of new technologies in space transport and in the mining sector to bring about a better future. Such celebration of a highly polluting industry often pioneered by autocratic eccentric billionaires and happening at a time of global climate breakdown and rising social inequalities might seem paradoxical. To give a sense of proportion, a single launch of Space X’s Falcon 9 emits 116 tons of CO<sub>2</sub> in the 165 seconds of its first launch stage.<sup>154</sup>

<sup>148</sup> Greason and Bennett, “Economics of Space,” 7–9. For an analysis of the historical interplay between US and Chinese rare-earth production see Sulgiye Park, Cameron L. Tracy, and Rodney C. Ewing, “Reimagining US Rare Earth Production: Domestic Failures and the Decline of US Rare Earth Production Dominance—Lessons Learned and Recommendations,” *Resources Policy* 85/A (2023), 104022, <https://doi.org/10.1016/j.resourpol.2023.104022>.

<sup>149</sup> Dunstan, “Space Barons.”

<sup>150</sup> Block and Huebert, “Environmentalists in Outer Space.” The idea of space as an “ideal waste-disposal site” (filled with hot suns acting like “giant incinerators”) was already put forward by Patrick Cox in a 1985 *Reason* article. See Patrick Cox, “Space Entrepreneurs.”

<sup>151</sup> Block and Huebert.

<sup>152</sup> Block and Huebert; Greason and Bennett, “Economics of Space,” 9–10; Dunstan, “Space Barons”; Littlewood, “True Spirit.”

<sup>153</sup> Greason and Bennett, “Economics of Space,” 10.

<sup>154</sup> Mark Piesing, “The Pollution Caused by Rocket Launches,” *BBC Future*, July 2022, at [www.bbc.com/future/article/20220713-how-to-make-rocket-launches-less-polluting](https://www.bbc.com/future/article/20220713-how-to-make-rocket-launches-less-polluting).

To justify their leap of faith, think-tankers emphasize the importance of uncertainty, thereby redeploying Hayek's mid-century framework based on the limits of human knowledge.<sup>155</sup> For instance, Pooley and Tupy's Simonian argument is also predicated on humanity's inability to know "the full extent of our resources," which encourages us, individually, to find ways to overcome these potential limits.<sup>156</sup> Saul Zimet (FEE) explicitly refers to Hayek when arguing that "the implications of future knowledge, by their very nature, are often unimaginable in advance."<sup>157</sup> As such, as Barthold (TechFreedom) and Greason and Bennett remark, we cannot predict the ways and forms future technological developments will take, as "[t]he possibilities in the future ... are limitless."<sup>158</sup> Zimet observes that, however "unpredictable the effects of new knowledge," they are likely to change the world "in ways that benefit virtually the entire human population, not just the privileged and wealthy."<sup>159</sup>

Hayek's theory of knowledge denounced planners' hubristic ambition to rationally organize and impose the "conscious social control" of the future.<sup>160</sup> In the same way, the think-tankers' stance about the uncertainty of the future is directed against (in Block and Huebert's words) the "deep ecology" of "state environmentalists."<sup>161</sup> For instance, Steele dismisses the IPCC report by arguing that it is "driven by ideology, not science."<sup>162</sup> True science would consider the economic cost of each of the policies advocated, acknowledging that interventionist policies might be more costly than what they are supposed to cure. As Zimet further argues, radical ecological policies "rarely facto[r] in ... that continued economic growth, facilitated in large part by fossil fuels, will likely continue to produce unpredictable technological and scientific breakthroughs, creating new forms of security and wellbeing, and at new scales."<sup>163</sup> Politicians should therefore not attempt to rein in industries based on their current environmental costs. Littlewood similarly notes that we should even restrain from criticizing what may appear as futile a venture as commercial space travel as it may result in unexpected "enormous benefits to wealth, health and Earth's environment" by making access to space resources cheaper.<sup>164</sup> Zimet defends space tourism with a similar argument, explicitly appealing to Hayek's celebration of the "extravagance and even waste" of the wealthiest few as opening the "path of advance": they will fund what seems at first a luxury, contributing to bringing its price down in the long run, thus making it affordable for the many.<sup>165</sup>

<sup>155</sup> Friedrich Hayek, "The Use of Knowledge in Society," *American Economic Review* 35/4 (1945), 519–30.

<sup>156</sup> Pooley and Tupy, "Simon Abundance Index."

<sup>157</sup> Zimet, "Elon Musk's Critics."

<sup>158</sup> Barthold, "Cripple Elon Musk"; Greason and Bennett, "Economics of Space," 13.

<sup>159</sup> Zimet, "Elon Musk's Critics."

<sup>160</sup> Friedrich Hayek, *The Road to Serfdom* (London, 2011), 38.

<sup>161</sup> Block and Huebert, "Environmentalists in Outer Space."

<sup>162</sup> Steele, "Climate Doom Ahead?"

<sup>163</sup> Saul Zimet, "Top Global Energy Agency Calls for Phasing Out of All Gas-Powered Cars, Coal-Fired Plants," *FEE Stories*, July 2021, at <https://fee.org/articles/top-global-energy-agency-calls-for-phasing-out-of-all-gas-powered-cars-coal-fired-plants>.

<sup>164</sup> Littlewood, "True Spirit."

<sup>165</sup> Zimet, "Elon Musk's Critics." See also Friedrich Hayek, *The Constitution of Liberty*, ed. Ronald Hamowy (Chicago, 2011), 98.

Such a rationale ends up flirting with some particularly problematic accelerationist positions, like Lilico's, who invites his *Daily Telegraph* readers to "[c]hoose between spending \$3-trillion on preventing the Earth heating up by 3C [*sic*] over the next 150 years and spending it on making Mars a blue planet"—arguing that the latter is more economically efficient.<sup>166</sup> For Lilico, it might ultimately make more economical sense to let the planet burn if it gives industries, and especially the most polluting ones like space manufacturing, the space they need to innovate, as they might stumble upon the technology that will save us all in the future. As he bluntly writes elsewhere,

in a century's time, perhaps our great grandchildren will be sitting in their climate-controlled domes on Mars thinking "The deal was this: London flooded, the Mississippi overflowed even more than usual, and in exchange we got the cure for cancer, interplanetary travel, a life expectancy of 200, eyelid head-up-displays, telepathy, and all the other apparatus of modern life. That was a no-brainer."<sup>167</sup>

Humanity's salvation might come from space, even if it entails a back-up composed of a select few sent to live on other planets (including Musk's Mars colony)—as Nelson and Block candidly imagined in their nuclear-annihilation scenario.<sup>168</sup> The neoliberal space fantasy here reveals itself in its most nihilistic dimension. It disinvests from and devaluates the present terrestrial environment in the name of a future utopia beyond current material limits.<sup>169</sup> A highly speculative future, that might never materialize, potentially abandoning the rest of the world to its apocalyptic fate.

## Conclusion

This article has shown that US and UK twenty-first-century neoliberals see space as an opportunity to reactivate their long-term struggle against collectivism, but also as a way to suture a number of epistemological threats to neoliberal thought such as the legitimization of private property or its unflinching support for capitalism.

Entrepreneurs like Elon Musk play a central role in this picture. Most of the writers analyzed do not hide their admiration for Silicon Valley's space barons and often publicly support them. With a few notable exceptions, their ideological defense of the free market accommodates itself surprisingly well with the tech billionaires' monopolistic tendencies.<sup>170</sup> Why? Think-tankers' eagerness to rush to Musk and his friends'

<sup>166</sup> Lilico, "The Red Planet." The American Enterprise Institute's website has also a dedicated section called "Faster, Please." See Benjamin Noys, *Malign Velocities: Acceleration and Capitalism* (Winchester, 2014); Smith and Burrows, "Software."

<sup>167</sup> Andrew Lilico, "Things to Believe and to Disbelieve about Climate Change," *Conservative Home*, Dec. 2007, available at <https://web.archive.org/web/20250522192625/https://conservativehome.blogs.com/columnists/2007/12/things-to-belie.html>.

<sup>168</sup> Nelson and Block, *Space Capitalism*, 4. In an interesting parallel, Musk justifies his Martian ambition by arguing that it will "enable us to back-up the biosphere"; Elon Musk, "Making Life Multiplanetary," *Twitter*, 5 June 2022, <https://twitter.com/elonmusk/status/1533410745429413888>.

<sup>169</sup> Melinda Cooper, *Life as a Surplus* (Seattle, 2008), 60.

<sup>170</sup> As a rare example, the arch-libertarian Nelson and Block critique Musk, Bezos, and Branson for their reliance on state contracts; Nelson and Block, *Space Capitalism*, 188–97.

defense can be explained by the important part given to the entrepreneur in the neoliberal imaginary—some neoliberal thinkers, such as former MPS president Herbert Giersch, even defend “innovative monopolistic competition” against “sterile perfect competition.”<sup>171</sup> Simon’s theories on the infinity of natural resources are, for instance, predicated on human ingenuity, and specifically on the ability of certain individuals to find alternatives to, or new ways of producing, resources in tension. Indeed, human creativity is the “ultimate resource” mentioned in Simon’s book title. It will enable humanity to bypass any present limits on physical resources. Thanks to it, “we and our descendants can manipulate the elements in such fashion that we can have all the raw materials that we desire at prices even smaller relative to other goods and our total income”; “our cornucopia is the human mind and heart.”<sup>172</sup>

Crucially, neoliberal think-tankers wholeheartedly integrate Simon’s celebration of the entrepreneur into their case for the marketization of space. For instance, Szóka emphasizes that “Julian Simon was right: the ultimate resource really is human ingenuity, and the resources of space are really only there for what we make of them.”<sup>173</sup> Space resources are effectively barren if not transformed by the visionary terraforming action of an Elon Musk. As Nelson and Block write, if private developers were left in charge of the Moon, they would create “a veritable Garden of Eden [that] inspires the imagination.”<sup>174</sup> A think tank like TechFreedom is thus dedicated not simply to promoting private space exploitation but, very explicitly, as its official documents state, to supporting the interests of entrepreneurs in order to “unleas[h] the ultimate resource: human ingenuity.”<sup>175</sup>

The neoliberal think tanks’ speculative plans for the future rely on a blind faith in the power of entrepreneurship to solve tomorrow the problems of the present. As Zimet writes, “Without visionaries like Elon Musk to bet their capital on the exploration of new technological frontiers, humanity’s future would be much less bright. But with them, there are no limits in view.”<sup>176</sup> The entrepreneur figure sustains the neoliberal discourse about the future by hiding its most blatant shortcomings: the lack of sustainability of and disruptions caused by the current capitalist modes of production. No need to curb today’s carbon emissions as tomorrow’s entrepreneurs shall find economically efficient (but energy-hungry) geo-engineered solutions to environmental problems.<sup>177</sup> Entrepreneurs like Musk, Bezos, and Thiel are so important to the neoliberal project because they convey in their discourse and embody in their practice an epic of space enterprise, which offers an opening on a hypothetical tomorrow that

<sup>171</sup> Giersch as quoted in Dieter Plehwe, “Schumpeter Revival? How Neoliberals Revised the Image of the Entrepreneur,” in Plehwe, Slobodian, and Mirowski, *Nine Lives of Neoliberalism*, 120–42, at 137. As Plehwe analyzes, while the entrepreneurial figure is part of a longer capitalist tradition, it has been given a central place in neoliberal thought, with the coexistence of diverse, and at times contradictory, theories of entrepreneurship.

<sup>172</sup> Simon, *The Ultimate Resource*, 67.

<sup>173</sup> Robert Zubrin and Berin Szóka, “The Future of Space,” Cato policy report (Jan.–Feb. 2020), at [www.cato.org/policy-report/january/february-2020/future-space](http://www.cato.org/policy-report/january/february-2020/future-space).

<sup>174</sup> Nelson and Block, *Space Capitalism*, 66–7.

<sup>175</sup> Dunstan, “Comments.”

<sup>176</sup> Zimet, “Elon Musk’s Critics.”

<sup>177</sup> Mirowski, *Never Let a Serious Crisis Go to Waste*, 340.

helps to displace the need for a robust decarbonization and energy transition. Their promising dreams of future off-world colonies provide the neoliberal think-tankers' speculative plans with additional reality and affective texture. Thanks to them, space, the final frontier, seems within humanity's reach.

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